SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 50**
Question Count: 10

Code of Conduct
I Athavan K(192524036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Athavan K

Assessment Tasks Class Test

1. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
2. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
3. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
4. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decisionmaking.
5. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

CO4-AT1

CLASS Test

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cloud computing
for Big Data
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CSA1807

1 Impact of Data Quality Issues in Large Scale analytics Platforms:

Data Quality plays a major role in the reliability of analytics results. In a Large scale Analytics platform inconsistency, duplication and missing values can significantly affect decision making.

Inconsistency occurs when the same data is represented differently across different sources. For ex, a customers gender may be stored as male, M and 1. This can produce incorrect analysis and reduce data accuracy.

Duplication occurs when the same record appears multiple times. Duplicate customer or Transaction records can lead to incorrect calculation, such as inflated sales figures (or) inaccurate customer counts.

Missing values :- occurs when important fields are not available. Missing customer age, location, transaction amount, or product information can reduce the effectiveness of analytical models and machine learning algorithms.

Scalable Recommendation system for an Online streaming service :-

A streaming recommendation system recommends movie (or) shows based on user perferences and behaviour. The system can collect watch history, ratings, likes, searches, viewing duration and skipped content.

The architecture can use Hadoop for distributed processing and cloud (or) distributed storage for storing large amounts of user data. Recommendation techinque such as collaborative filtering and content based filtering can be used.

User behaviour analytics helps identify user interests. For example, if a user frequently watches action movies, the system can recommed similar movies. Distributed processing allows millions of user records to be processed simultaneously, making the system scalable.

8 Role of cloud computing in Big Data Ecosystems:

cloud computing provides scalable ~~information~~ for infrastructure for storing and processing large amounts of big data. It provides services such as cloud storage, databases, distributed computing, analytics and ML.

Advantage:

(i) Scalability :- Resources can be increased (or) decreased according to demand

(ii) Cost management : Pay as you use models ~~include~~ reduce the need for expensive hardware.

(iii) Flexibility : Organizations can quickly deploy different big data service

(iv) Availability : cloud infrastructure provides backup and redundancy.

Limitations :

(i) un controlled usage can increase costs

(ii) security and privacy risks may occur

(iii) Organizations may become dependent on a cloud provider

(iv) Network problems can affect performance.

9 Real time Analytics Framework for Transportation and Logistics :-

A real time transportation analytics system collects data from GPS devices, IOT sensors, vehicles, traffic systems, and delivery systems.

Technologies such as Apache kafka and spark streaming can process his data in real time.

The system analyzes vehicle location, traffic condition, fuel consumption, speed and delivery status, Real time traffic data can be used to identify the best routes and avoid congestion.

Big data technologies improve:

(i) Route optimization by analyzing traffic and historical route data

(ii) Fuel efficiency by identifying unnecessary travel and idling

(iii) Decision making by providing real-time vehicle and delivery information.

(iv) Maintenance by identifying abnormal vehicle behaviours

10 Ethical and legal challenges of Big Data :-

Big data creates several ethical and legal challenges. The major issues include privacy, security, lack of transparency, unauthorized data collection and algorithm bias.

Organization may collect personal information without users clearly understanding how it will be used. Large databases can also become targets for cyber attacks and data breaches. Biased data can produce unfair results in automated decision-making.

Organization should implement policies such as:

* obtain user consent before collecting personal data
* use encryption and access control to protect information
* collect only the data that is actually required
* Follow applicable data protection and privacy regulations
* regularly audit algorithm for bias and fairness
* clearly explain how user data is collected and used.